Chapter 6 exercises

Use real matrix entries unless specified otherwise. State stability conditions and the restrictions imposed by real initial data.

CH06-E1

For

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix},$$

1. Find eigenvalues from the characteristic polynomial.
2. Give a basis of each eigenspace and its geometric multiplicity.
3. Solve $u_{k+1} = Au_k$ for $u_0 = (2, 0)^T$.

CH06-E2

Let

$$A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}.$$

1. Find eigenvalues and eigenvectors.
2. Give $A = X\Lambda X^{-1}$.
3. Compute a closed form for A^5, rather than repeated multiplication as the final method.

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CH06-E3

Given

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

1. Find algebraic and geometric multiplicities.
2. Decide diagonalizability.
3. Use $A = 2I + N$ directly to compute A^k.

CH06-E4

Solve the initial-value system

$$\frac{du}{dt} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} u, \qquad u(0) = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

1. Find eigenvalues and orthonormal eigenvectors.
2. Project the initial vector onto both eigendirections.
3. Give $u(t)$ and its long-term behavior.

CH06-E5

Consider

$$\frac{du}{dt} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} u.$$

1. Find the complex eigenvalues and eigenvectors.
2. Give the real-valued general solution $u(t)$.
3. Prove $\|u(t)\|^2 = \|u(0)\|^2$, so each orbit stays on a circle.
4. If the matrix is replaced by

$$\begin{bmatrix} \alpha & -1 \\ 1 & \alpha \end{bmatrix},$$

characterize stability as α varies.

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CH06-E6

Let

$$S = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

1. Check symmetry without first expanding the characteristic polynomial.
2. Prove every eigenvalue is real.
3. Prove eigenvectors for distinct eigenvalues are orthogonal.
4. State the spectral theorem's conclusion and explain why eigenvectors cannot be chosen arbitrarily.

CH06-E7

Let

$$S = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

1. Test positive definiteness through eigenvalues, leading minors, and energy.
2. Express $x^T S x = 1$ in eigen-coordinates and give its semiaxis lengths.
3. Find the maximum and minimum of $x^T S x$ subject to $\|x\| = 1$.

CH06-E8

Let

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}.$$

1. Find the eigenvalues, trace, and determinant of A.
2. Compute $e^{At} = e^{-t} e^{Nt}$, where $A = -I + N$.
3. Determine asymptotic stability of the zero solution.
4. Explain why negative eigenvalue real parts do not in general imply monotone decrease of $\|e^{At}\|_2$. Use $A + A^T$ to determine whether this particular matrix is actually contractive. Give a different matrix and initial vector if needed to exhibit transient growth.

CH06-E9

1. For invertible square A, B, prove AB and BA have the same eigenvalues by similarity or determinants.
2. Prove a diagonalizable matrix with all eigenvalues zero is the zero matrix.
3. Give a nonzero 2×2 matrix with all eigenvalues zero and explain why it is not diagonalizable.

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CH06-E10

A nilpotent matrix N on a 10-dimensional space has $(d_0, d_1, d_2, d_3, d_4, d_5) = (0, 3, 6, 8, 9, 10)$, where $d_k = \dim \ker N^k$.

1. Find the counts of blocks with length at least k and exactly k, and give the Jordan form.
2. Find rank N, rank N^2, and the least positive m for which $N^m=0$. Explain why d_1 is not the largest block size.

CH06-E11

Let $B = \begin{bmatrix} -1 & 4 \\ 0 & -1 \end{bmatrix}$, $u_0 = (1,1)^T/\sqrt{2}$, and consider u′=Bu.

(1) Find $e^{Bt}u_0$ and prove asymptotic stability of the zero solution.

(2) Find the maximum norm for this initial state and its time, and compare with C=−I under the same initial condition.

(3) Use H=(B+B^T)/2 to explain the initial energy derivative. Find a nonzero initial state that decays monotonically under B. Distinguish the peak in part 2 from the matrix exponential's operator norm.

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CH06-E12

Let $A = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$. Find its characteristic polynomial and eigenspace dimensions.

Construct a Jordan chain and an invertible P such that the size-two block of $P^{-1}AP$ has 1 on its superdiagonal. Verify each chain relation.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 6.

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